

Project Title: Automatically classify behaviour of tracked objects to enhance the safety of autonomous systems.

Batch No. IOT-15
AY: 2025-2026 (PE-2)

Abstract

Our research focuses on in-cabin passenger safety, specifically autonomous cars without drivers. In such cases, medical emergencies may arise, and no one will be present to assist the passenger. To address this problem, we will implement and deploy an in-car safety system using a ZED2 camera, an ECG sensor, and deep learning models. The deep learning model YOLO is based on object detection. YOLO-Pose and Eye Blinking for performing tasks like YOLO-Pose for posture and Eye Blinking for eye state. ECG data is used to detect heart rate abnormalities. The proposed solution is to build a safety system for the car using cameras, ECG sensors, and a Nvidia Jetson SDK / Nvidia RTX-3050 to process and detect if any emergency. Automation will be triggered, causing the vehicle to redirect itself to the nearest hospital and send alerts using the communication protocol.

Objective

To track the passengers health using an ECG sensor and a live camera for continuous monitoring. Camera data is processed with the help of deep learning models such as YOLOv8, YOLOv8-Pose, and Eye Blinking to detect and analyse passenger behaviour. The fusion of camera and sensor data is implemented using the Jetson AGX SDK (HPC) / Nvidia GeForce RTX 3050 to enable real-time decision-making. To validate the system model, it will be integrated into our real buggy for live testing and deployment.

Design Strategies

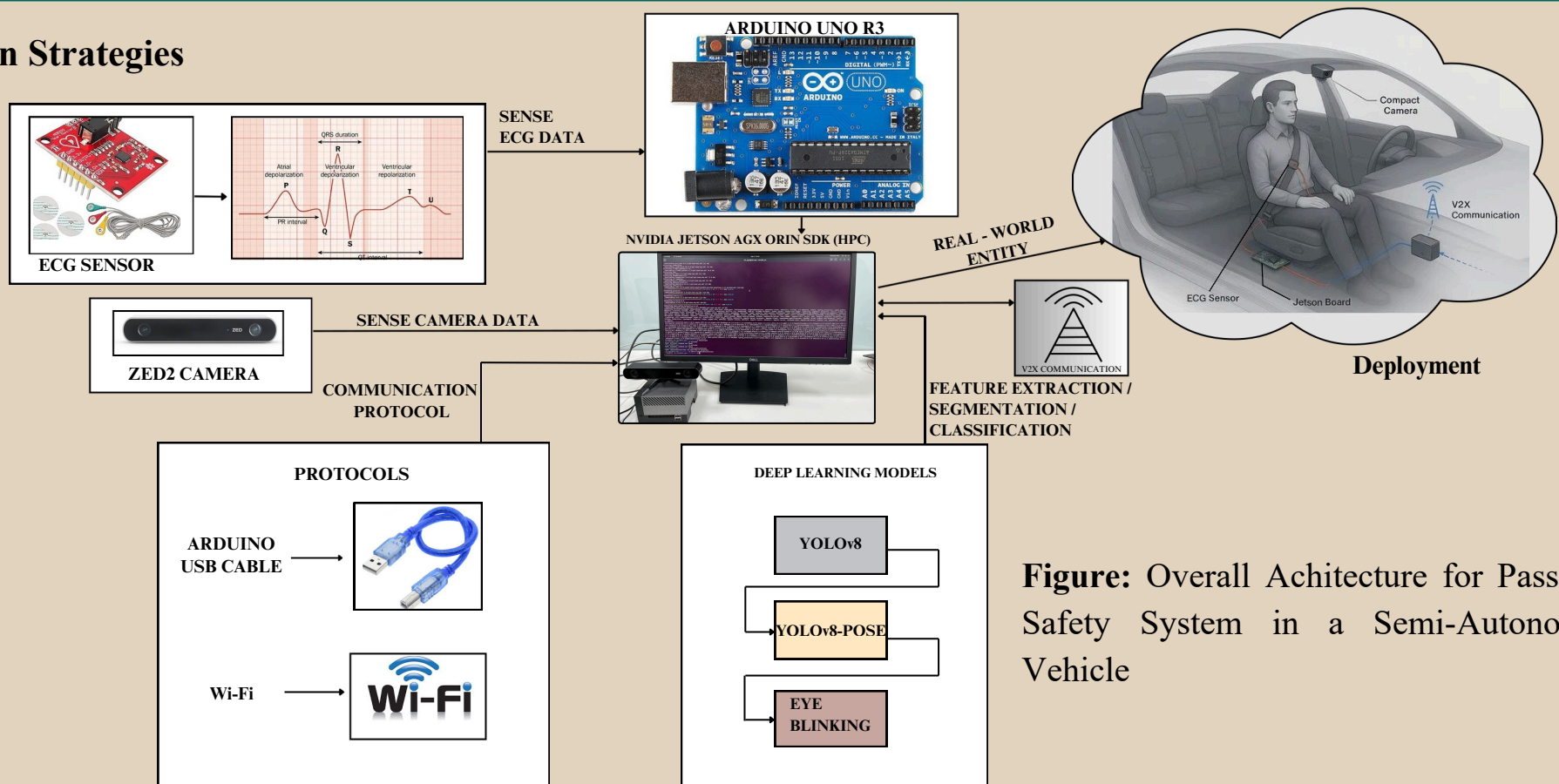


Figure: Overall Achitecture for Passenger Safety System in a Semi-Autonomous Vehicle

Results



Fig 1.1: Vehicle is localized at Starting node (sd_side).

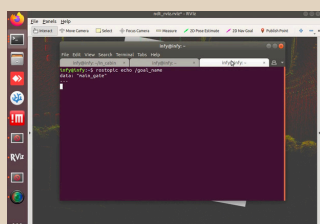


Fig 1.2: Vehicle assigned to Destination (main_gate).

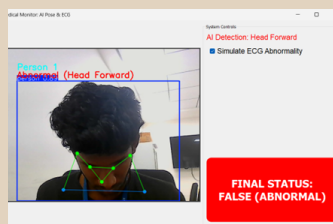


Fig 1.3: Both posture and heart rate shown as abnormal then, final state shown as False.



Fig 1.4: If abnormality detects as shown in Fig 1.3, then the vehicle will stop.

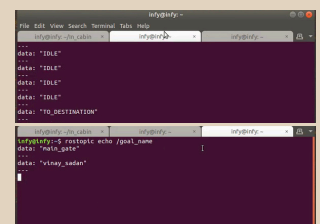


Fig 1.5 (a) & (b): Vehicle moving from IDLE state to Destination.



Fig 1.6: Vehicle is rerouted to the health care point (Vinay_sadan).

Identification of Technologies /Tools/Algorithms

Deep Learning(AI)
Edge Computing (Nvidia Jeston AGX Orin SDK / Nvidia GeForce RTX-3050)
Embedded System / IoT

Conclusion

Our project designed and implemented for health management system in autonomous vehicles to detect abnormality of a passenger using Camera and ECG monitoring. YOLOv8-Pose analyzes person's posture, and eye state, while an ECG sensor tracks heart rate. A multi-validation logic triggering emergency responses only when both posture and heart rate abnormalities occur. In such cases, a Boolean signal is sent via LAN to the vehicle, which stops vehicle safely and reroute to the nearest hospital.

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